

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Applied **data-driven methods**, including simulation and digital engineering, to enhance safety and reliability in rail operations.
- Contributed to concept and feasibility studies for new and existing **rail assets**, ensuring alignment with project goals and client requirements.
- Collaborated within **multi-disciplinary teams**, engaging with stakeholders and delivery partners to deliver integrated rail solutions.
- Developed Python scripts to automate data extraction and transformation for operational reporting, reducing manual effort by approximately **30%**.
- Managed production incidents for enterprise applications using **Splunk** and **Dynatrace**, resolving P1/P2 issues within defined SLAs.
- Supported system integrations through the development and maintenance of **Java Spring Boot REST APIs**, facilitating cross-system data flows.

SKILLS

Rail Domain – Rail Operations, Planning, Timetabling, Operational Modelling, Systems Engineering, Safety Assurance, Telecoms, Digital Systems, Digital Engineering, MEP, Project Management, Design Management.

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, Data Modelling, Data Warehousing, Statistical Analysis, R, RStudio, Python (pandas, NumPy, scikit-learn).

Software Engineering – Java, Spring Boot, Python, REST APIs, SQL, PostgreSQL, MySQL, Oracle, Linux, Git, Jenkins, Docker, Jira, Confluence.

Cloud & Observability – AWS (EC2, S3, Lambda, CloudWatch), Splunk, Dynatrace.

Methodologies – Agile, Scrum, Change Management, Requirements Gathering, UAT, Incident Response.

PROJECTS

Rail Network Performance Analysis Platform

- Developed a **Python-based data pipeline** to ingest and process historical train performance data, covering ~500,000 records for analysis.
- Built interactive **Power BI dashboards** to visualize key performance indicators, including punctuality and capacity utilization, for operational stakeholders.
- Integrated with **SQL databases** to extract and validate data, ensuring data integrity for reporting and simulation inputs.

Automated System Monitoring and Alerting Tool

- Engineered a **Java Spring Boot application** to collect system health metrics from various enterprise services.
- Implemented alerting mechanisms using **Splunk** integrations, reducing critical incident detection time by an estimated **25%**.
- Containerized the application using **Docker** for consistent deployment across development and production environments.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)